STOCK PRICE FORECASTING

USING ECONOMETRIC AND MACHINE LEARNING MODELS

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FINANCIAL ECONOMETRICS

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INTRODUCTION

Theoretical Background: Classical Econometric Models:

As mentioned above, before presenting my findings, it is worth to describe what the classical econometric models actually are. While in the case of machine learning models I am searching for hidden and complicated connections between features, classical models are based on a rather simple principle: the most reliable forecast of the next moment equals the previous one.

Below there is a brief description of each of three types of classical models I used to forecast Reliance and Vodafone Idea's performance:

1. Auto-Regressive (AR) Model:

The Concept: AR stands for "auto regressive", meaning self-predicting. Therefore, the model forecasts the future stock price only based on its previous values.

How it works: Imagine the weather forecasting: you expect tomorrow's climate conditions based on those which happened yesterday and the day before. If a stock's price has some momentum (meaning a good/bad day is followed by another good/bad one), AR model takes it into account.

In this Project: In the context of my research, AR(2) model means that the algorithm uses the information of exactly two previous days to calculate the price for day three.

2. Moving Average (MA) Model

The Concept: Unlike AR model, MA does not use previous prices of a particular stock, but instead, the previous forecasting error (randomness).

INTRODUCTION

How it works: Unexpected pieces of news about the company may lead to temporary rise/fall of the share price. The assumption is that the randomness will average in due course, so the forecast is based on how much it was wrong before.

In my Project: Using the moving average model I meant MA(2), so the algorithm used the forecasting errors made two days ago to create the forecast for tomorrow.

3. ARIMA (Auto-Regressive Integrated Moving Average):
The Concept: As the name implies, this type of models incorporates all aspects of AR and MA ones with an additional feature.

How it works: Both auto-regressive and moving average models are efficient when applied to stationary processes – that is, those that have constant mean and variance over the period. However, stock prices generally have trend (rise or fall). To make the time series stationary, I calculate the difference between consecutive prices. This operation is called integrated.

In my Project: Why did I consider ARIMA to be the most suitable classical model? As mentioned above, my data included some upward trend for Reliance. Thus, adding drift to ARIMA helped the algorithm take that momentum into account and give more accurate predictions.

ARIMA MODEL (IDEA)

In order to overcome the shortcomings of AR and MA models, several ARIMA models were developed for Vodafone Idea stocks. The different models tried were ARIMA(1,1,0), ARIMA(1,1,1), and ARIMA(2,1,1). The results obtained from AIC values indicated that the most suitable model was ARIMA(1,1,1).

Initially, the predictions using ARIMA remained almost constant, meaning that the model failed to capture the trend of the data. This was resolved by introducing a drift in the model.

The final model of ARIMA performed better than AR and MA models:

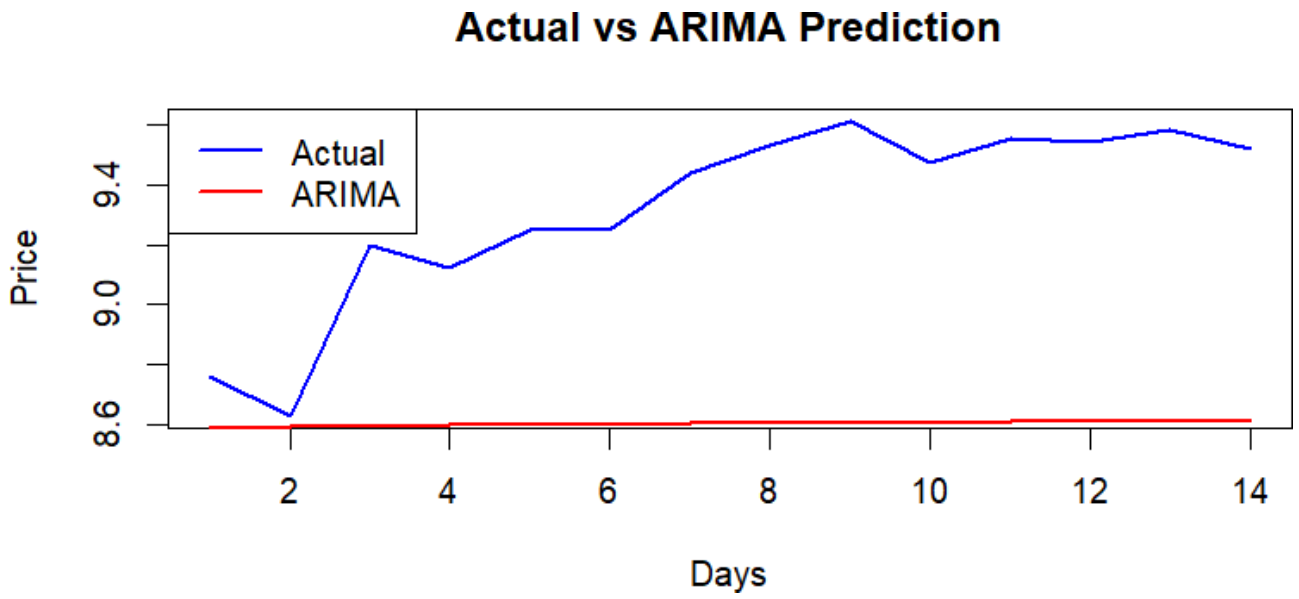
Model	MSE	MAE	MAPE
AR(2)	0.6297	0.7364	7.81%
MA(2)	0.6224	0.7315	7.75%
ARIMA(1,1,1)	0.5902	0.7117	7.54%

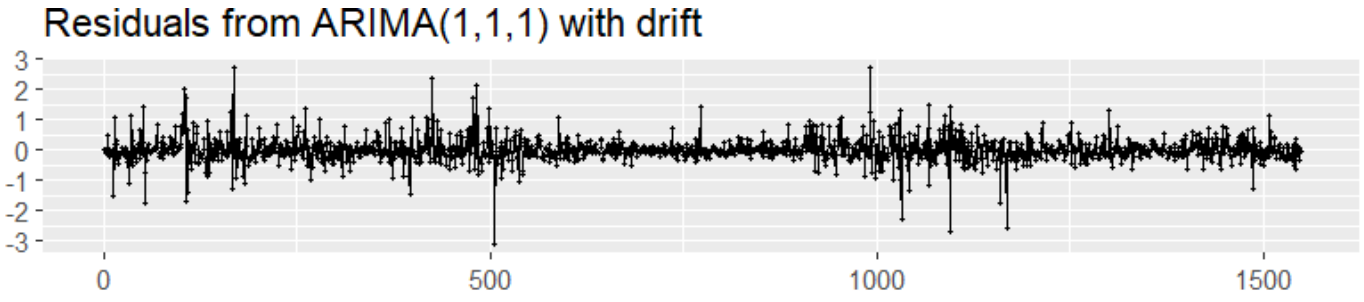
The validity of the model was proven through residual diagnostics showing that there is no autocorrelation (Ljung-Box test p-value> 0.05).

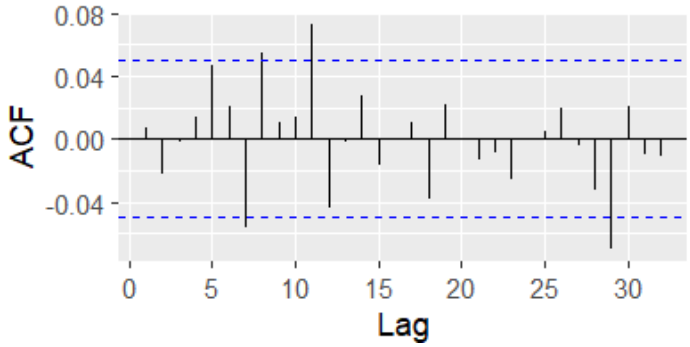
Nevertheless, the forecasts continue to look fairly flat relative to the real values, demonstrating how ARIMA fails to reflect the volatility and upward trend exhibited by the stocks.

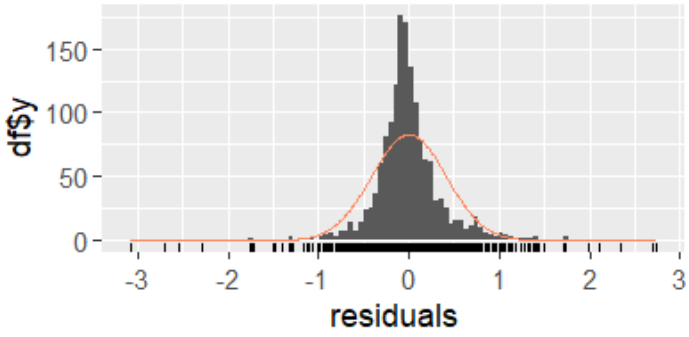
ARIMA MODEL (IDEA)

In general, even though ARIMA provides more accurate forecasts than AR and MA, it still falls short of modeling highly volatile data series such as stock prices.









ECONOMETRIC MODELING:
AR AND MA MODELS (IDEA)

Autoregressive and moving average models were applied for the stock price analysis of the company Vodafone Idea (IDEA.NS). Given that the initial time series was not stationary, a first-order difference was made to achieve stationarity.

Among the various models tested, the AR(2) and MA(2) models have been selected based on the AIC values.

These models were tested for a 14-day period using the following test data:

Model	MSE	MAE	MAPE
AR(2)	0.6297	0.7364	7.81%
MA(2)	0.6224	0.7315	7.75%

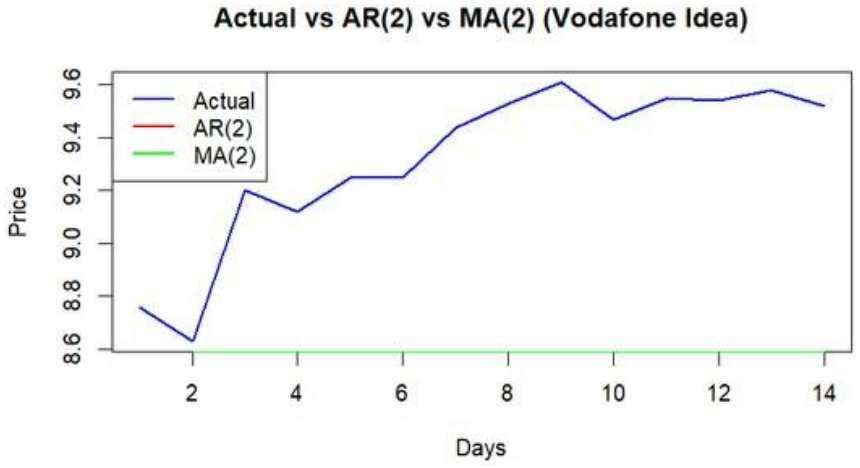
While the MA(2) model works slightly better than the AR(2) model, the distinction is minor.

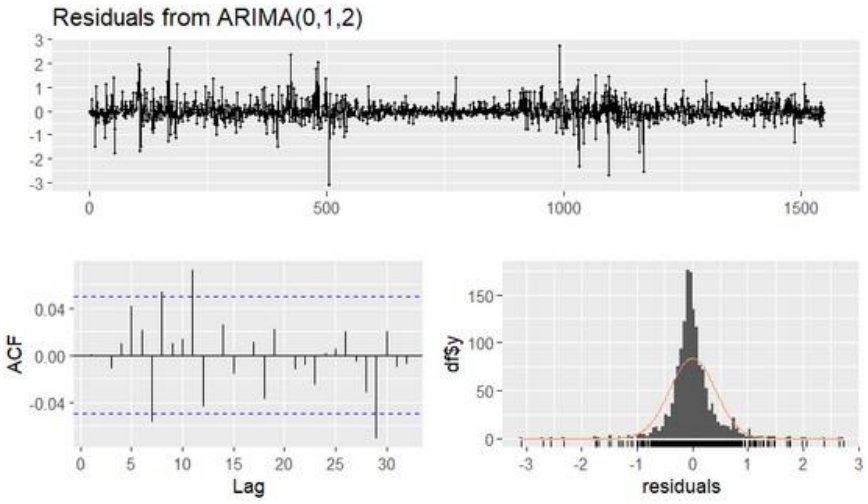
Model verification was performed through residual analysis. In the Ljung-Box test results, the p-values were found to be 0.071 for AR(2) and 0.075 for MA(2). As both exceed 0.05, the residuals lack any notable autocorrelation and can be considered white noise.

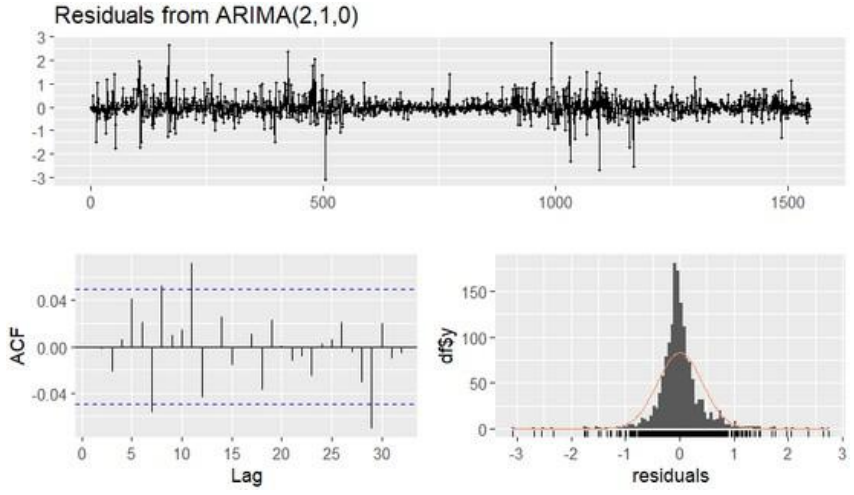
Nevertheless, when the forecasted values are compared to real data, both models create nearly constant predictions for the 14-day period, which suggests that neither model is capable of capturing trends and volatility in the stock prices.

ECONOMETRIC MODELING:
AR AND MA MODELS (IDEA)

To conclude, while the models AR(2) and MA(2) have proven to be statistically valid, they are not adequate for forecasting the Vodafone Idea stocks' dynamics. Thus, it is necessary to use a more sophisticated model, such as ARIMA.







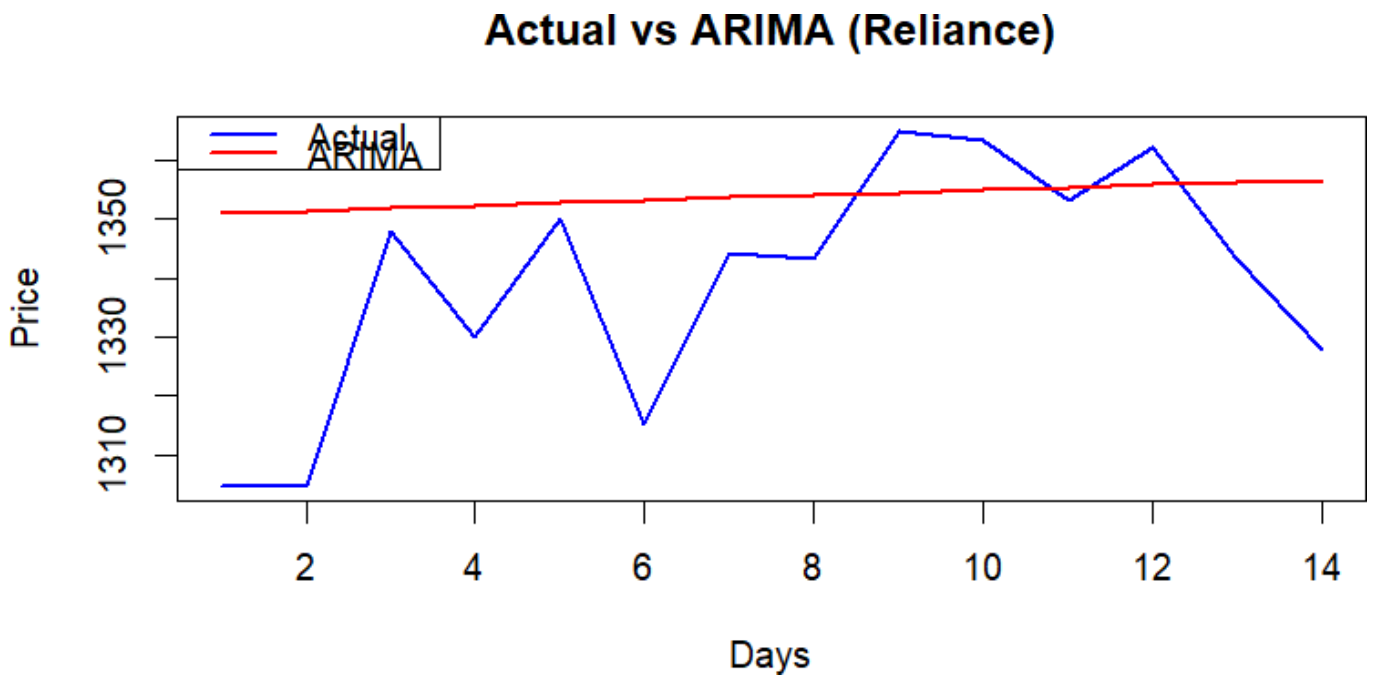
ARIMA MODEL (RELIANCE)

To account for trend, a drift term was included in the model. The resulting forecasts showed a slight upward movement, aligning reasonably well with the observed stock behavior.

The model achieved strong performance metrics:

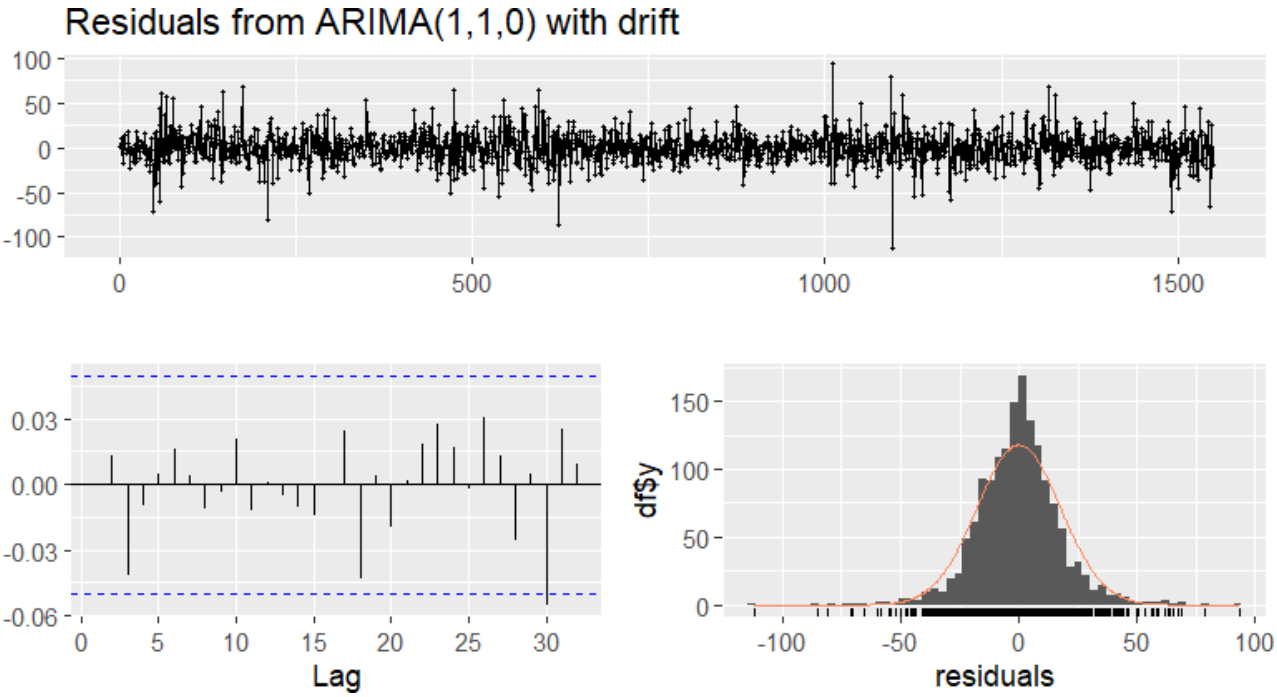
- MSE: 553.47
- MAE: 17.83
- MAPE: 1.34%

The low MAPE value indicates high forecasting accuracy.
Residual diagnostics confirmed that the model is statistically valid. The residuals are centered around zero, show no significant autocorrelation, and follow an approximately normal distribution. The Ljung-Box test further supports that residuals behave like white noise.



ARIMA MODEL (RELIANCE)

Compared to the Vodafone Idea stock, the ARIMA model performs significantly better for Reliance due to the relatively stable and less volatile nature of the data.
Overall, ARIMA(1,1,0) provides an effective and reliable model for forecasting Reliance stock prices.



ECONOMETRIC MODELING:
AR AND MA MODELS (RELIANCE)

On Reliance stock data, Auto-Regressive (AR) and Moving Average (MA) models were fitted following first-order differencing for stationarity.
Several models were considered based on combinations of p and q values for AR(p) and MA(q) models. As per AIC and BIC measures, it can be seen that AR(1) and MA(1) models are the best choices as they gave the smallest AIC values compared to other models.

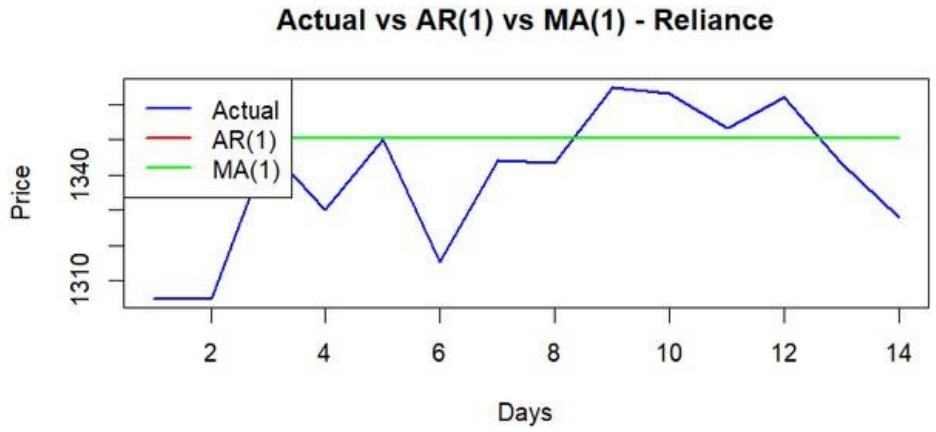
A residual analysis was conducted to validate the chosen models. The following characteristics of residuals were observed from the plots:

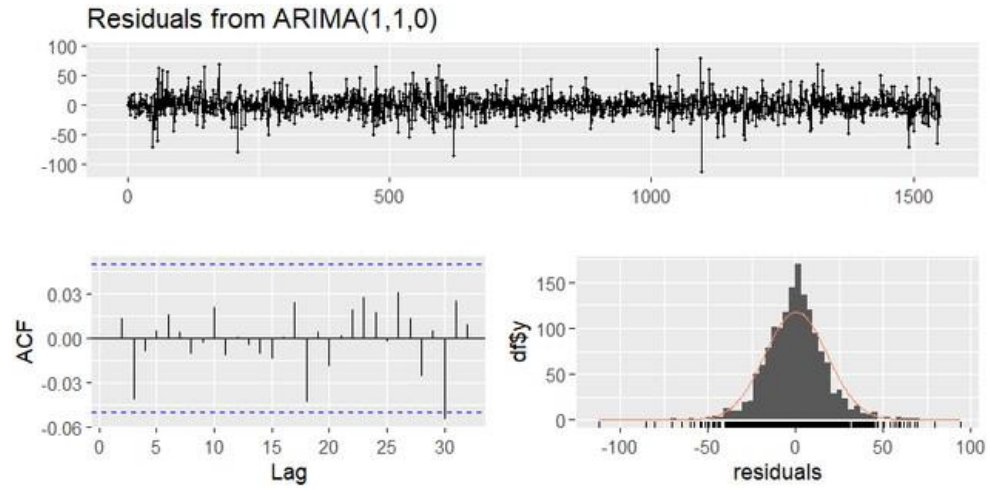
- Centered around zero
- No autocorrelation
- Normally distributed

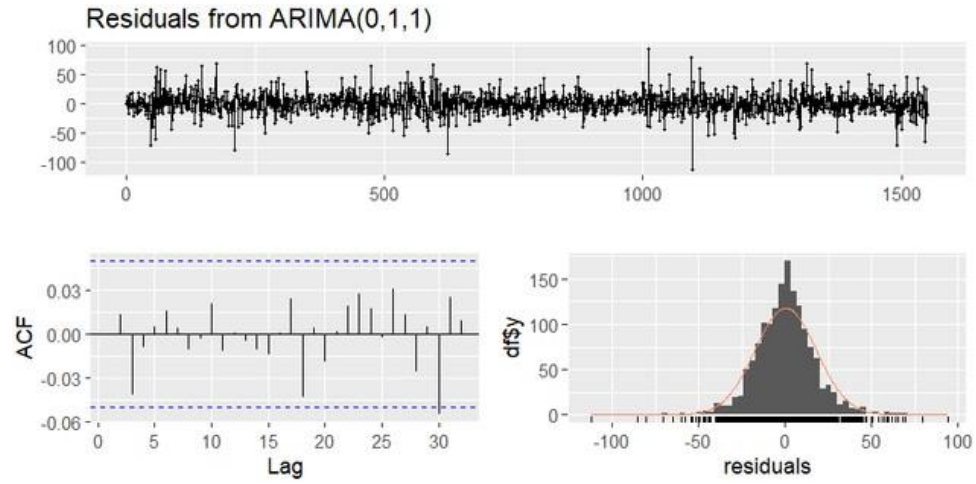
Further, by conducting Ljung-Box tests on residuals, it can be concluded that the residuals follow white noise with a p-value greater than 0.05.

When it comes to forecasting, both models yielded good results with very little difference in their performances. This indicates that the stock price series does not have much volatility, hence simple models could work well.
From the above discussion, it can be inferred that both AR(1) and MA(1) models are excellent baseline models for Reliance stock series.

ECONOMETRIC MODELING:
AR AND MA MODELS (RELIANCE)







COMPARISON PART FOR ARMA AND ARIMA

- **Comparative Analysis: Vodafone Idea vs Reliance**
- Econometric models' efficiency can be estimated through comparative analysis. Two stocks – Vodafone Idea and Reliance – were chosen for the analysis. While Vodafone Idea is highly volatile, the second one is relatively stable.

- **AR and MA Model Comparison**
- Vodafone Idea's optimal models are AR(2) and MA(2), which correspond to the lowest AIC values. Despite achieving low error metrics, they do not demonstrate fluctuations or upward trends in stock prices. Thus, AR and MA models cannot efficiently work with highly volatile stocks.
- The optimal models for Reliance are AR(1) and MA(1). They demonstrate relatively smooth predictions close to actual data. Besides, the AIC values of both models are very close, indicating that the stock behaves predictably.

- **ARIMA Model Comparison**
- ARIMA(1,1,1) was selected for Vodafone Idea. While it demonstrates lower AIC and errors compared to AR and MA models, its forecast still is nearly constant. It fails to show changes in the dataset.

COMPARISON PART FOR ARMA AND ARIMA



- As for Reliance, ARIMA (1,1,0) model was used since it has a minimum AIC. Drift has been introduced to capture an upward trend. Thus, forecasting performance is improved by this feature. The ARIMA model yields an excellent accuracy (low MAPE ~1.34%).

COMPARISON PART FOR ARMA AND ARIMA

- **Important Findings :** Vodafone Idea is a very volatile time series; hence, it is difficult to capture it using AR, MA and ARIMA models. On the other hand, Reliance shows relatively stable behavior, and thus, even basic models such as AR(1), MA(1) can work effectively. ARIMA helps improve the forecasting results in both cases but works better for a stable data type. The performance of the econometric models depends highly on the type of data analyzed

- **Conclusion:** The above comparison vividly shows that classical econometric models work well on stable and organized datasets (as in the case of Reliance). In contrast, they show poor performance on volatile time series (like that of Vodafone Idea).
- This highlights that model selection is not universal but depends on the statistical properties of the dataset.
- Therefore, ML models will be more appropriate.

MACHINE LEARNING APPROACH

- **My Model Selections :**

For my machine learning experiment, I selected four models to forecast the daily closing price of the stocks of RELIANCE.NS and IDEA.NS:

- **Random Forest and XGBoost:** Both belong to tree-based models.
- **Support Vector Regression (SVR):** It is a regression model that attempts to construct a model where the target value lies between two parallel hyperplanes.
- **LSTM (Long Short-Term Memory):** It is a neural network that is especially constructed to deal with the time series nature of data.

I trained all four of the above models using my historical stock data to make predictions for the next 14 trading days.

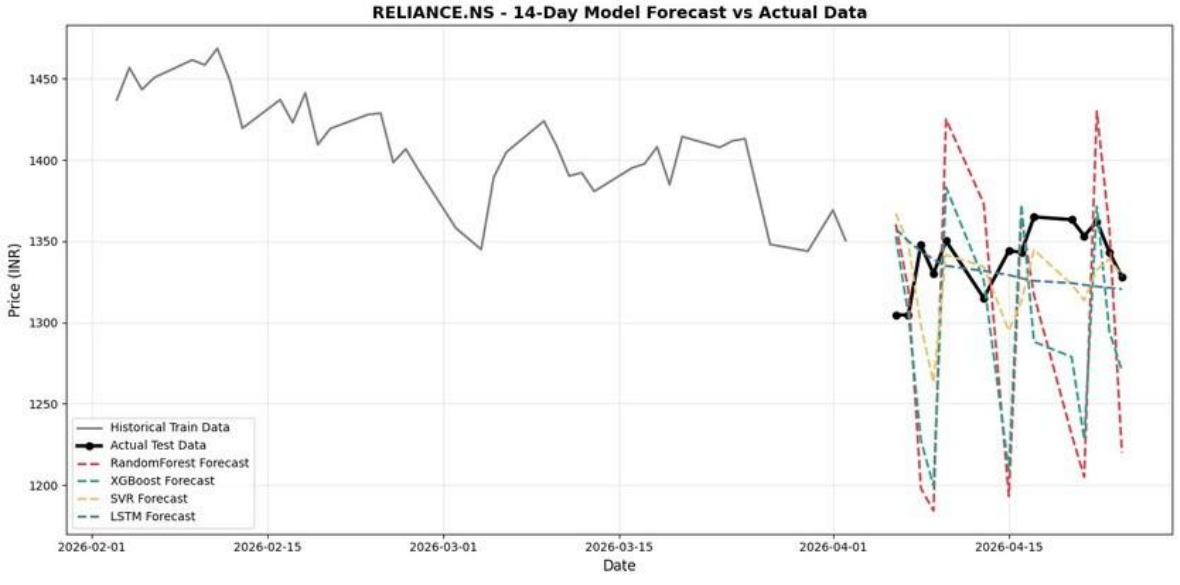
- **Error Analysis**

In order to find out the best-performing model, I used three well-known error measurements such as RMSE, MAE, and MAPE. In other words, the smaller the value is, the better the results were.

1. ERROR ANALYSIS FOR RELIANCE.NS:

RandomForest: RMSE: 99.1403 | MAE: 85.1173 | MAPE: 6.34%
XGBoost: RMSE: 79.5994 | MAE: 65.0342 | MAPE: 4.84%
SVR: RMSE: 38.7283 | MAE: 33.1709 | MAPE: 2.48%
LSTM: RMSE: 29.3614 | MAE: 25.1347 | MAPE: 1.88

->BEST PERFORMING MODEL: LSTM (Lowest RMSE: 29.3614)



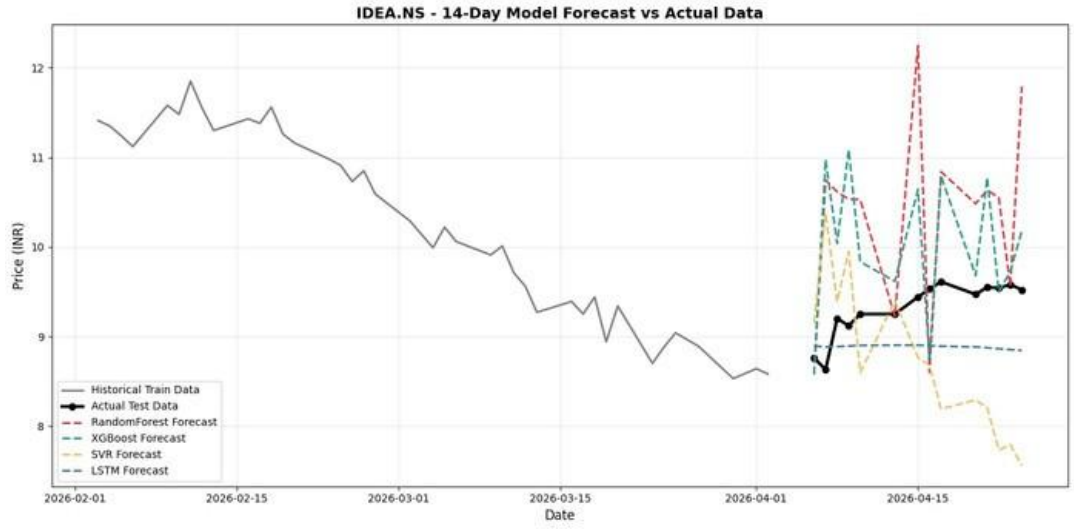
MACHINE LEARNING APPROACH

- **Error Analysis**

2. ERROR ANALYSIS FOR IDEA.NS:

RandomForest: RMSE: 1.4339 | MAE: 1.1999 | MAPE: 12.90%
XGBoost: RMSE: 1.0704 | MAE: 0.8383 | MAPE: 9.10%
SVR: RMSE: 1.2281 | MAE: 1.0711 | MAPE: 11.45%
LSTM: RMSE: 0.5298 | MAE: 0.4900 | MAPE: 5.20%

->BEST PERFORMING MODEL: LSTM (Lowest RMSE: 0.5298)



- **Examining the Graphs :**

As soon as I examined my forecast charts, together with the error results, several interesting things became apparent:

Random Forest and XGBoost: The forecast curves for these models were highly aggressive. As it is visible from the charts, these lines are very volatile, as they seemed to memorize random noise from the previous stock movements rather than identifying the market pattern. That is why these models resulted in the largest errors.

MACHINE LEARNING APPROACH

Support Vector Regression (SVR): As mentioned above, SVR performed relatively well when forecasting the prices for the Reliance company (the error rate was only 2.48%). Nevertheless, this model faced many difficulties when predicting the prices for the Vodafone Idea company. During the last few days of the 14-day period, the forecasts dropped sharply while real prices remained relatively stable.

Long Short-Term Memory (LSTM): This model proved to be quite efficient. As can be seen from the charts, the LSTM forecasted values (blue dotted lines) are more smooth compared to the rest. Moreover, this model was able to identify the trend direction properly.

- **Conclusion to Machine Learning:**

  From the numbers of errors and graphs generated above, LSTM can be considered the best machine learning algorithm for prediction. This is because it has the lowest number of errors for both stocks; Reliance with an RMSE of 29.36 and Vodafone Idea with an RMSE of 0.53. Since LSTM is built for sequence retention, it was the best suited for the short-term predictions of stocks.

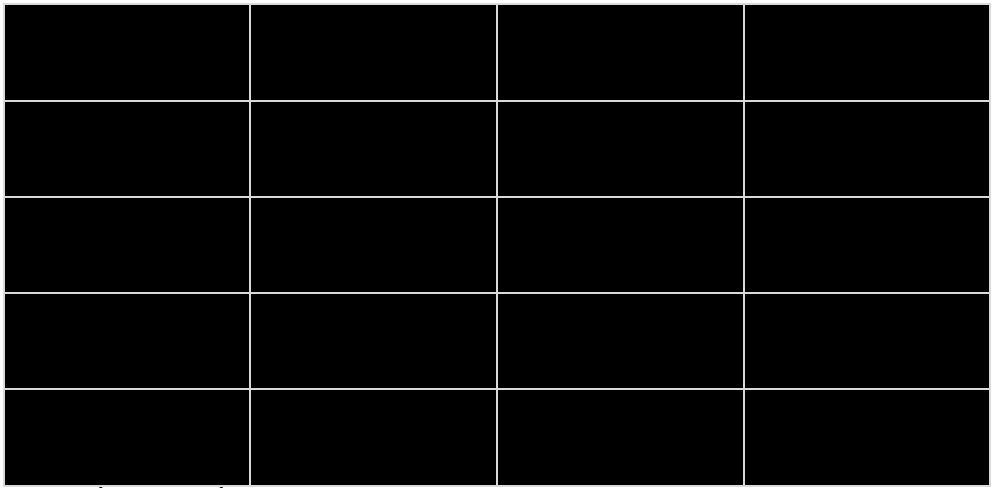
COMPARATIVE ANALYSIS

- **The Showdown:**

  In the concluding part of the current project, I decided to test my two best models – the best classical econometrics model and the best machine learning model – against each other by comparing their accuracy in forecasting stock prices.

  Our winner in Part 1 is ARIMA (ARIMA(1,1,0) for Reliance and ARIMA(1,1,1) for Vodafone Idea). As for Part 2, I will use my winning model, an LSTM neural network.

- **Final Model Comparison (ARIMA vs LSTM):**



- **Analysis for Reliance (The Stable Stock):**

  Upon analyzing the Reliance dataset which is generally quite stable, predictable and orderly, it came to my surprise that it is the conventional econometrics model that proved superior to the modern machine learning one.

  In fact, ARIMA gave a surprisingly low MAPE of only 1.34% compared to LSTM's 1.88%. It became possible after including a "drift" parameter in the ARIMA model which helped it to precisely detect the upward trend of the stock price. Consequently, I came to the conclusion that whenever a dataset is not noisy, tidy and orderly, it might happen that the traditional statistical models may provide better accuracy than modern AI.

COMPARATIVE ANALYSIS

- **Analysis for Vodafone Idea (The Volatile Stock):**

Everything dramatically changed with the analysis of Vodafone Idea. It is quite volatile and unpredictable stock, and its behavior is very challenging for the simple ARIMA model. As was mentioned before in Part 1, the result of forecasting with the help of the ARIMA model became completely unconvincing: it drew almost a straight line, thus making MAPE 7.54%.

However, Machine Learning Model performed better on such dataset. It was able to take into account the volatility of the dataset, and therefore MAPE went down to 5.20%.

- **Final Conclusion :**

It can be concluded from the above comparison that there is no such thing as an all-purpose “best” method for predicting stock prices; it is purely dependent upon the nature of the data itself.

In case the stock data is very stable and shows a definite trend (e.g., Reliance), conventional econometrics techniques such as ARIMA are extremely efficient and highly accurate.

In case the stock data is very volatile and erratic (e.g., Vodafone Idea), classical methods will fail miserably, and I need to use the advanced ML approach of LSTM networks.